

It's Always DNS ***So Let's Break It On Purpose***

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DNS in 5G Core

5G

Specific Private Zones

Domains used in Mobile Network only such as 3gppnetwork.org



Moderate Throughput

10s to 100s of requests/second



Low Latency

DNS is an important factor in the overall performance of the Mobile Network



Requirements for the 5G Core DNS Service



Geo Redundant & HA

Spread across multiple K8s clusters and geo regions to increase reliability

X No singletons



Fully Automated

GitOps driven and automated provisioning of DNS records

X No manual interaction allowed



Proximity to Consumer

Minimal amount of hops between 5G Core and DNS

X No SaaS allowed



Minimal Amount of SPOFs

Share nothing by removing single points of failure from the System

X No shared mgmt system



K8s integration with ExternalDNS

The System leverages Kubernetes Patterns such as CRs and Operators

X No CRUD outside kube-api



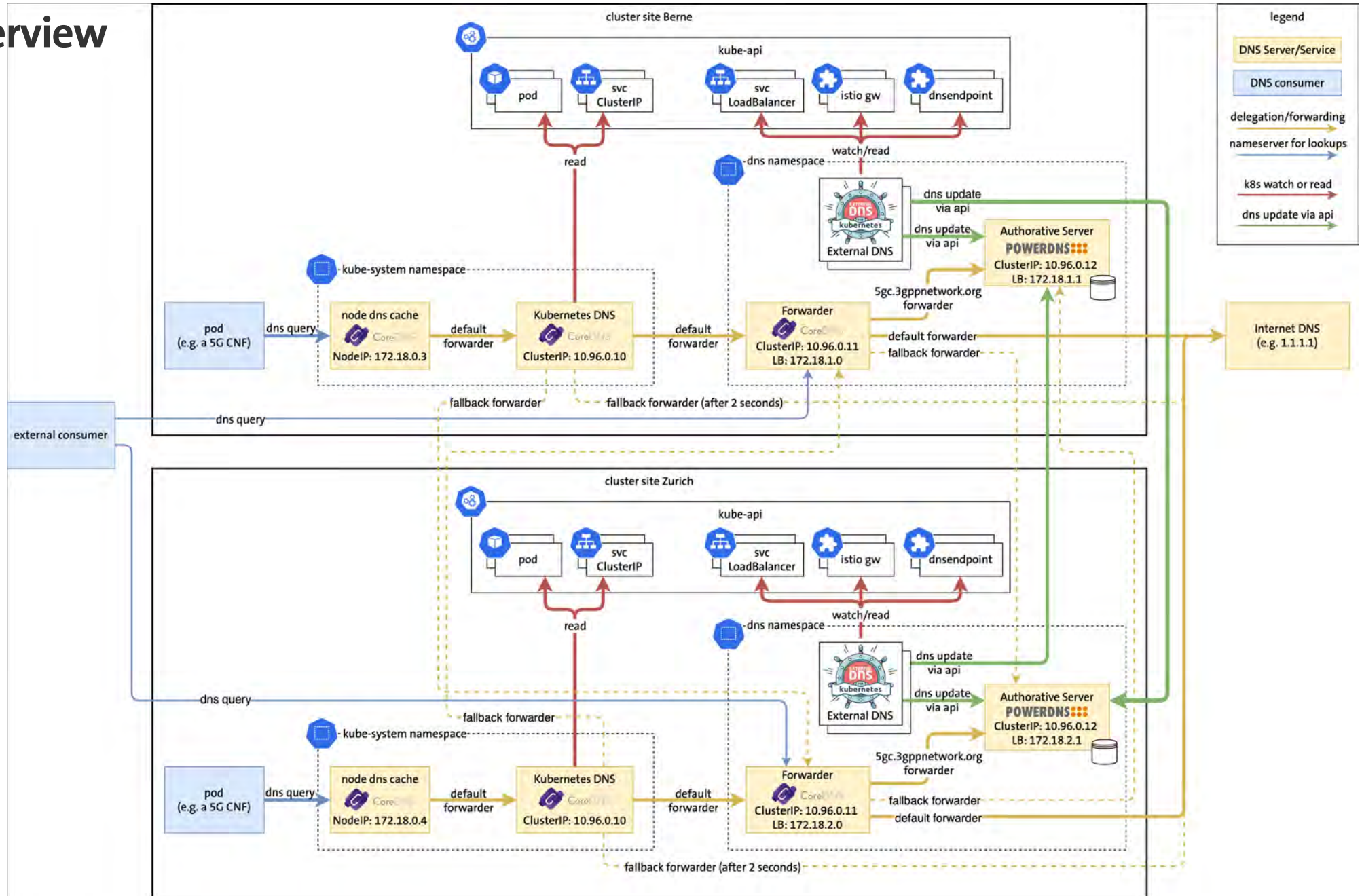
Support of Advanced DNS features

Resource Records such as NAPTR and SRV supported for e.g. SIP Phone Calls

X Need to go beyond A and CNAME

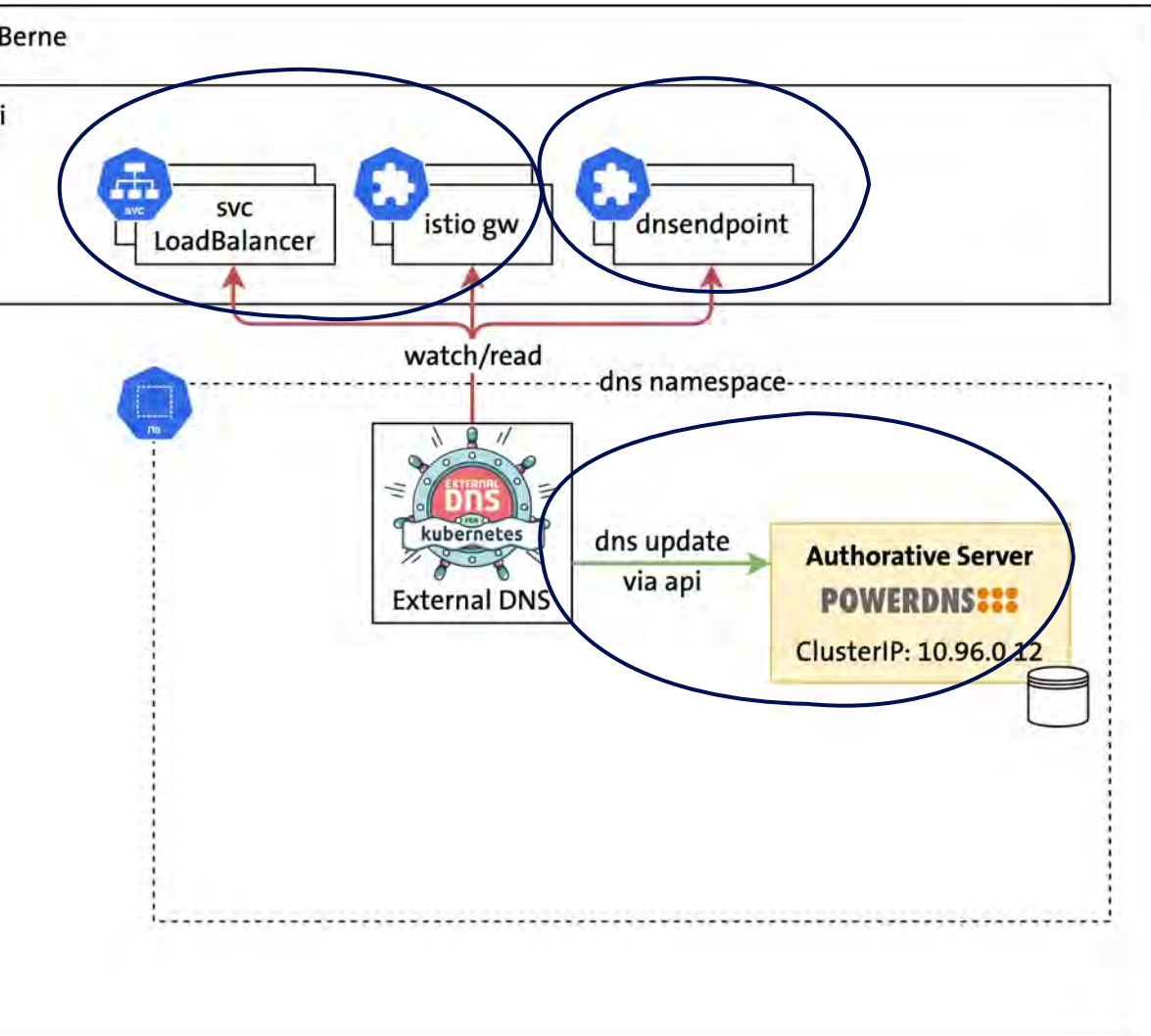


Overview





Automation of Authoritative Server Using ExternalDNS



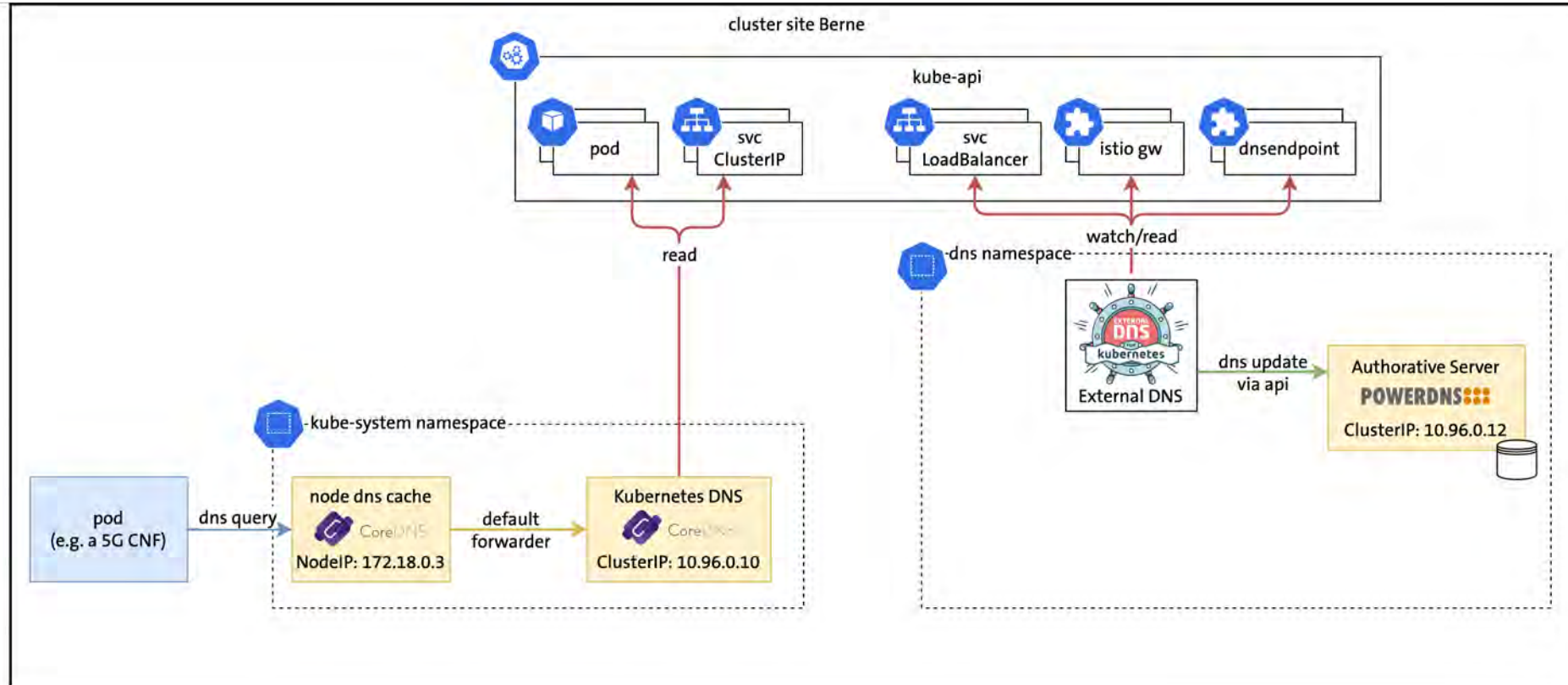
ExternalDNS reads from kube-api:

- Static Resource Records as DNSEndpoint Custom Resources
- Dynamic Type A Records using Annotations
- Name definition via Annotation
- IP fetched from Service / Ingress / Istiogw status field

ExternalDNS writes to PowerDNS API

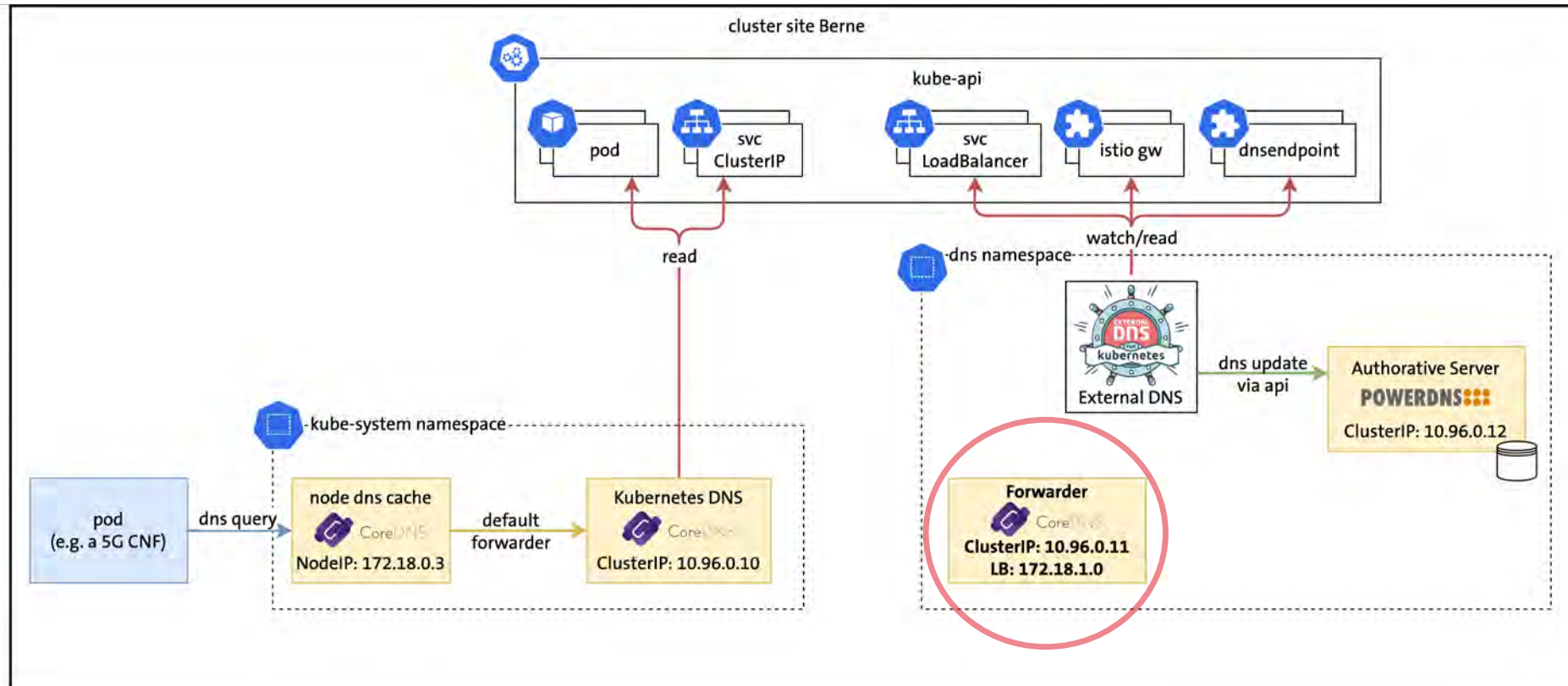


Forwarding to Authoritative Server



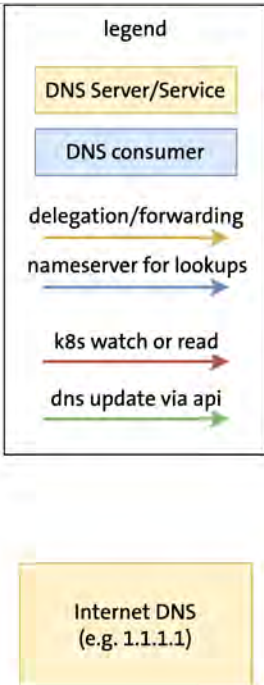
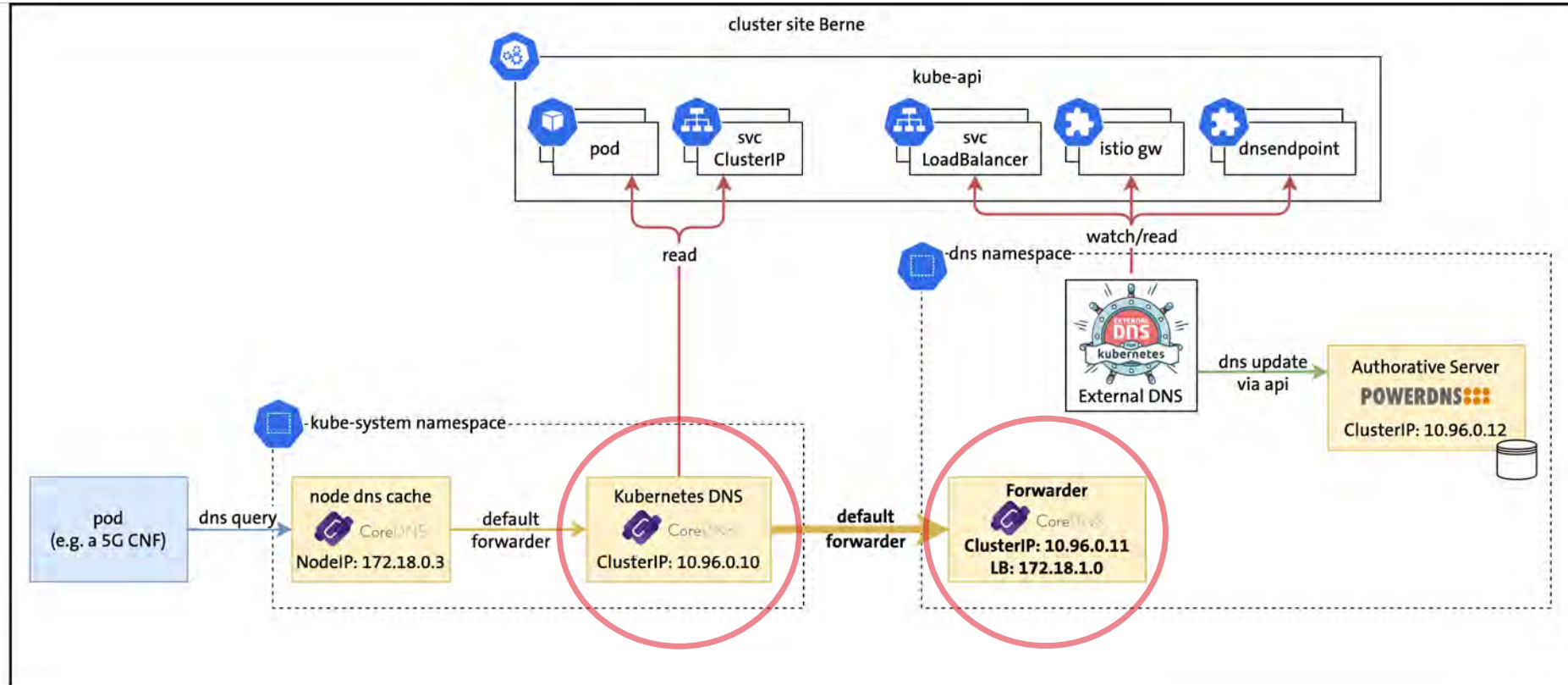


Forwarding to Authoritative Server



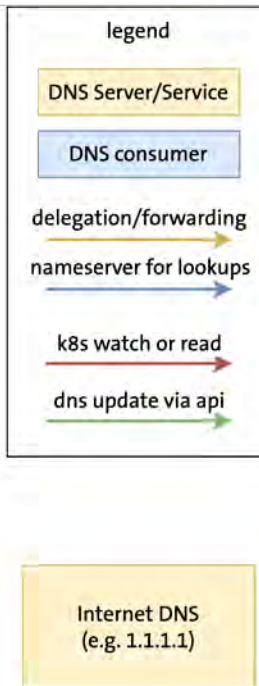
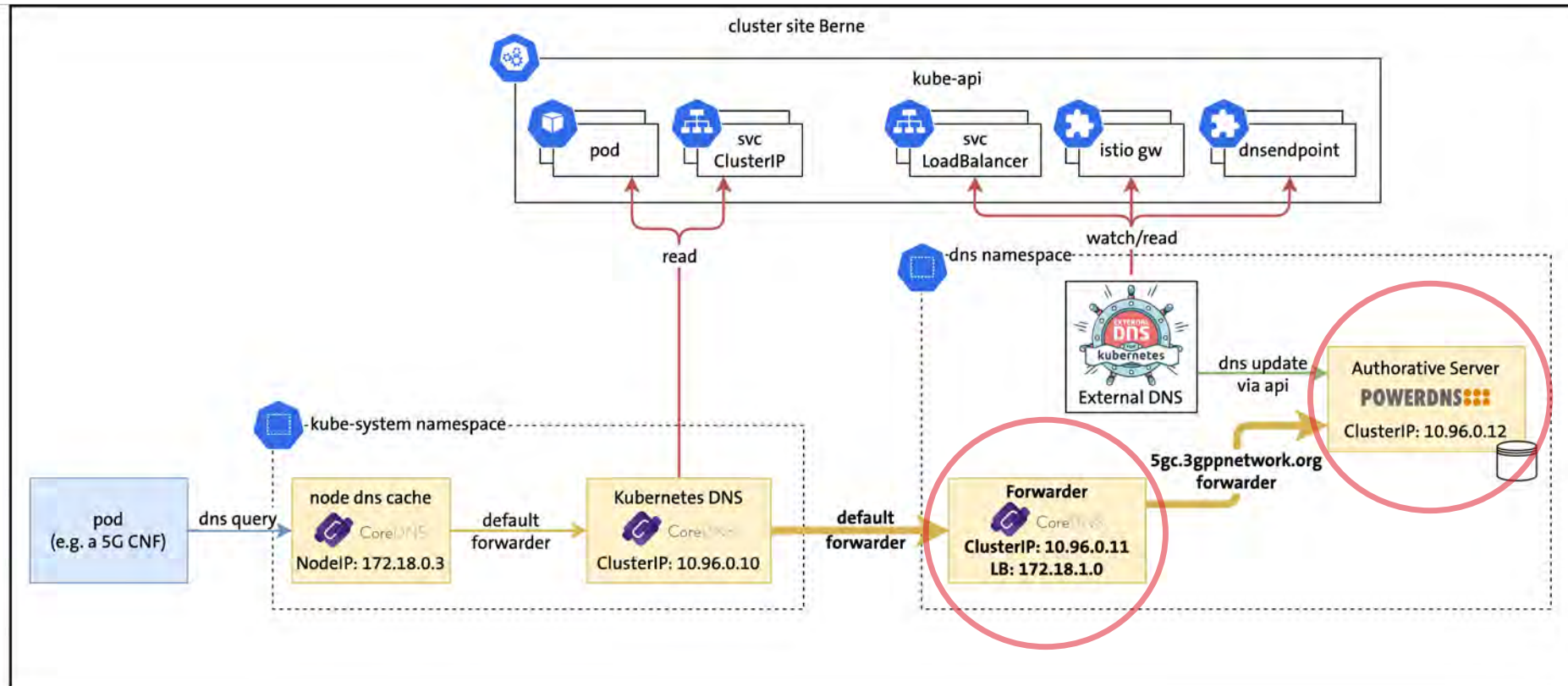


Forwarding to Authoritative Server



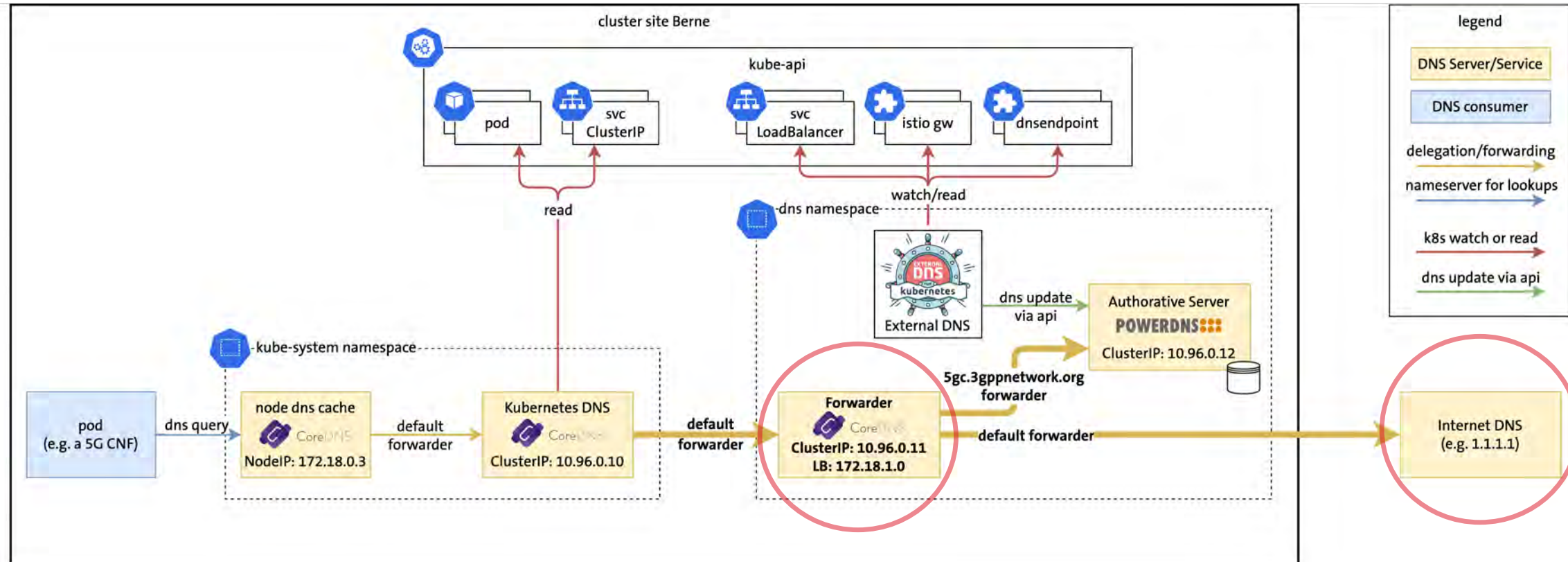


Forwarding to Authoritative Server



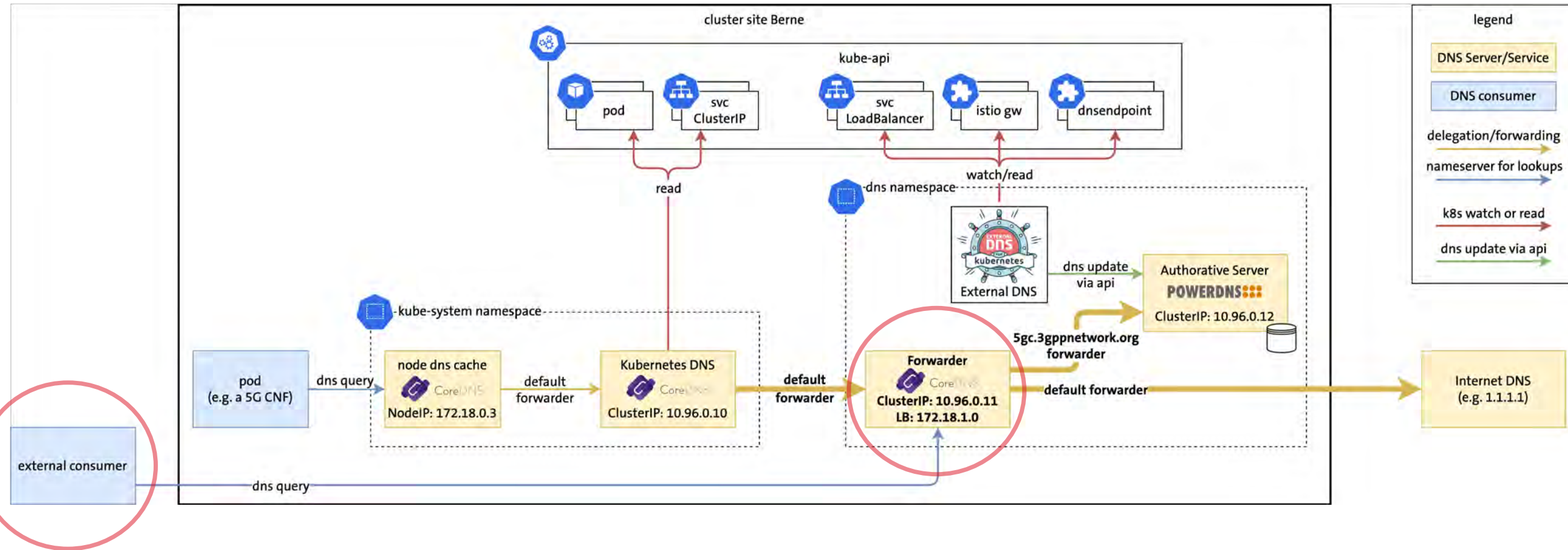


Forwarding to Authoritative Server



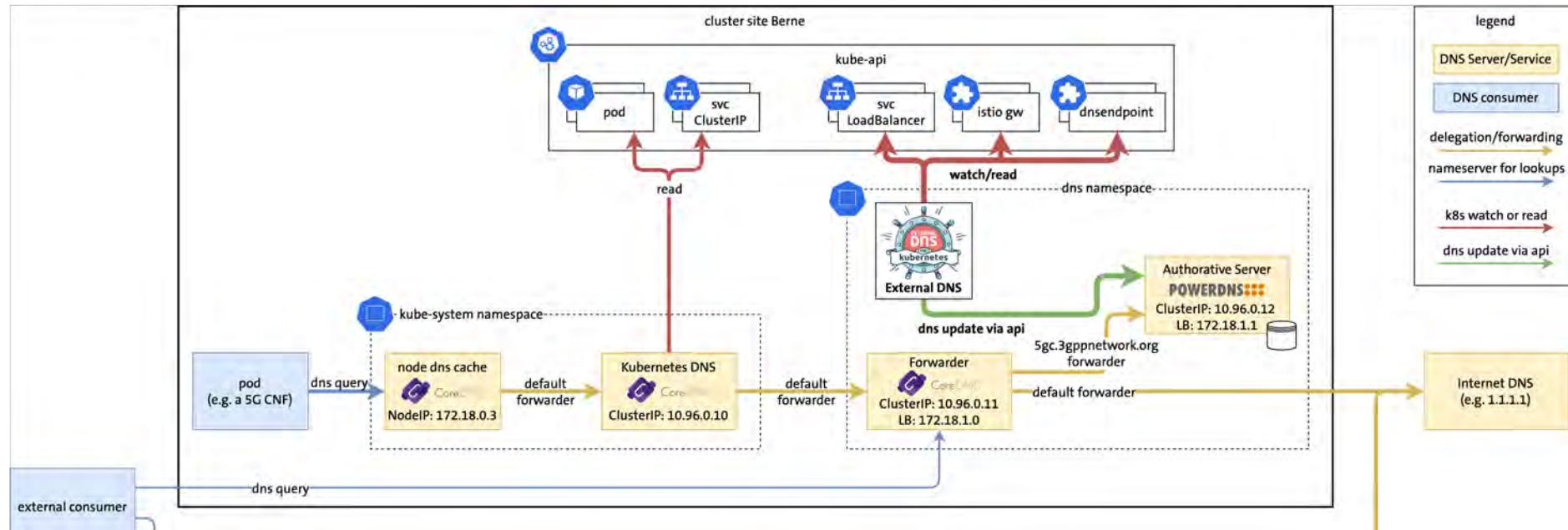


Forwarding to Authoritative Server



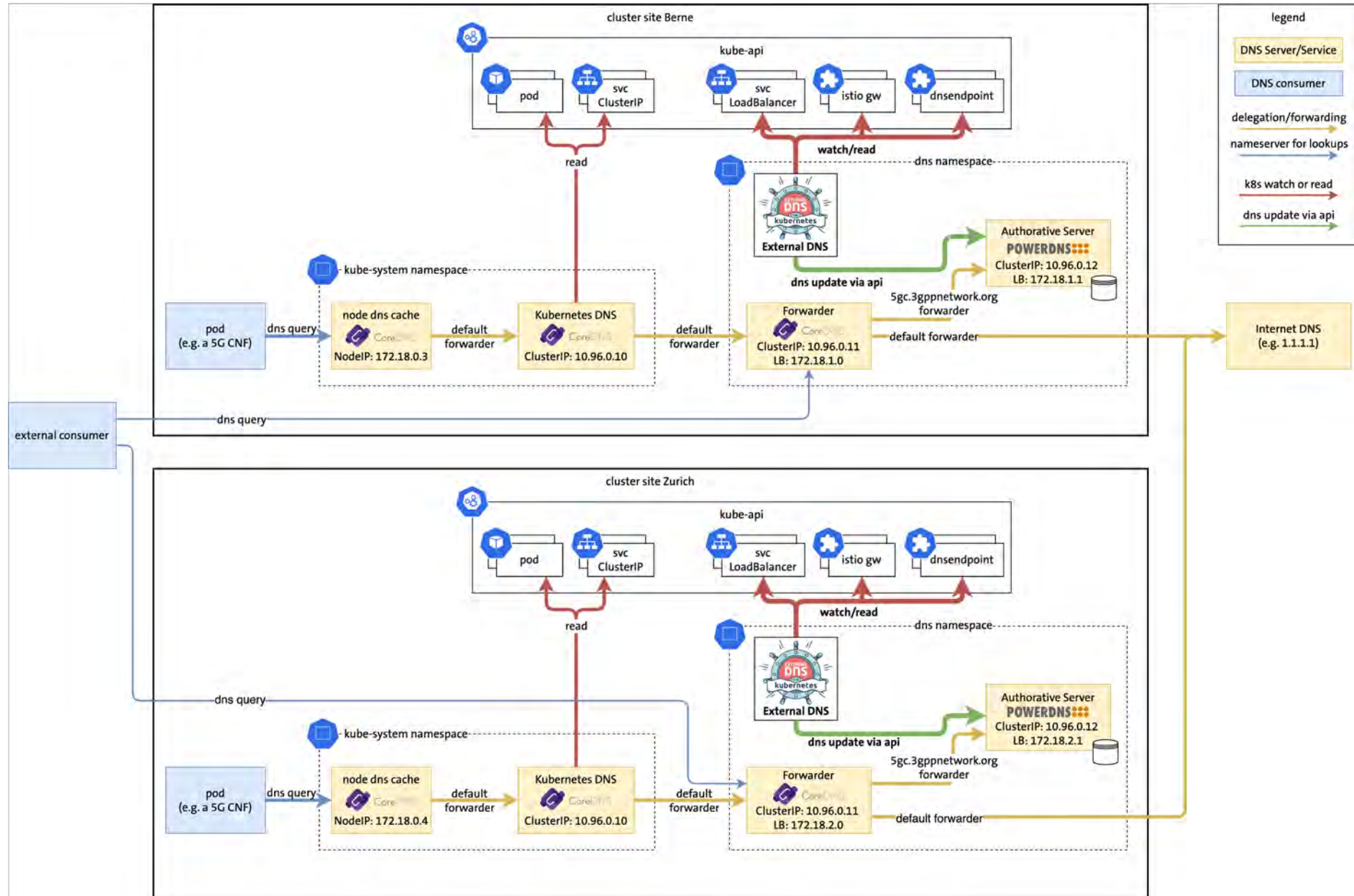


Dual-Cluster Using ExternalDNS



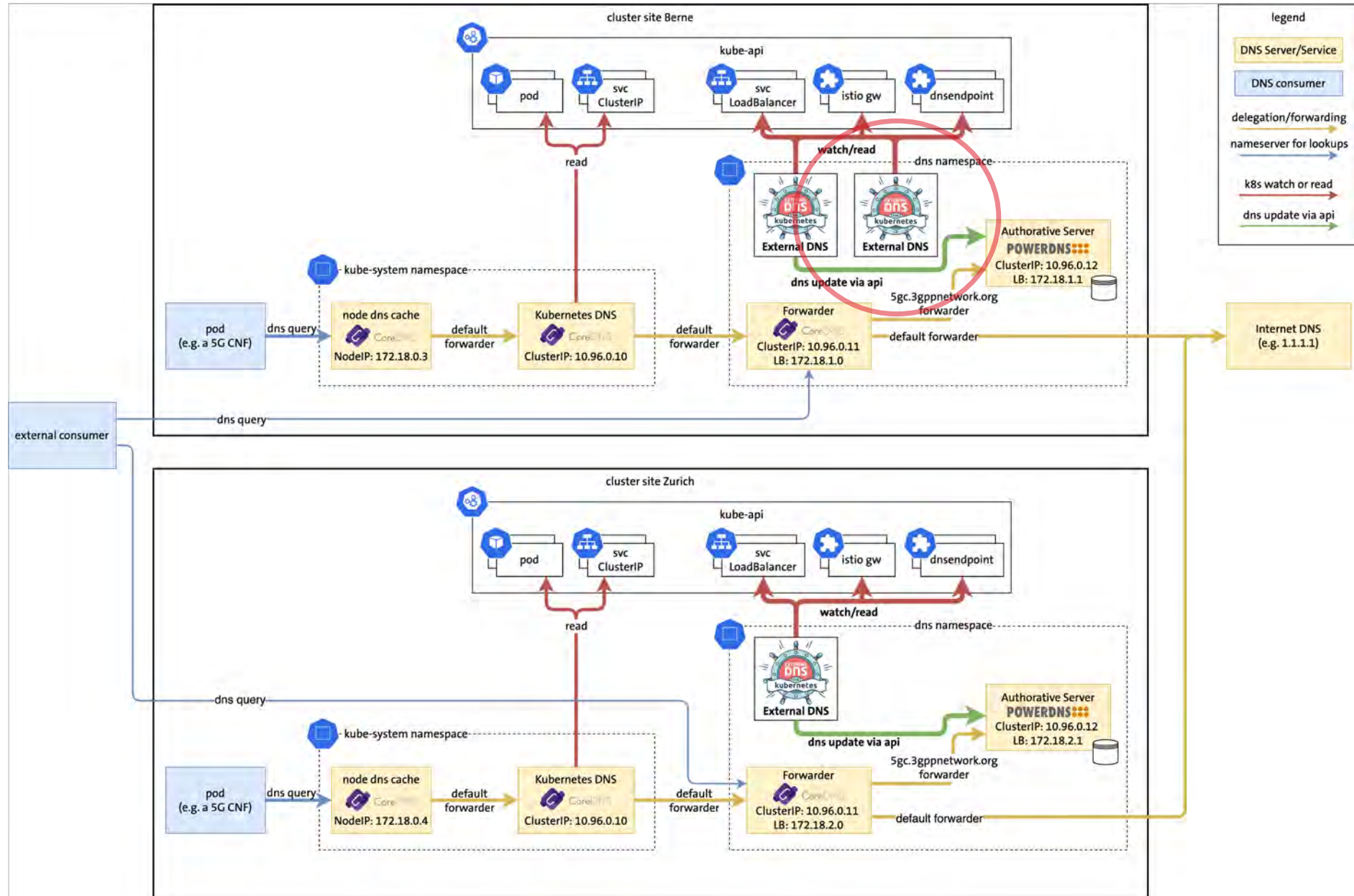


Dual-Cluster Using ExternalDNS



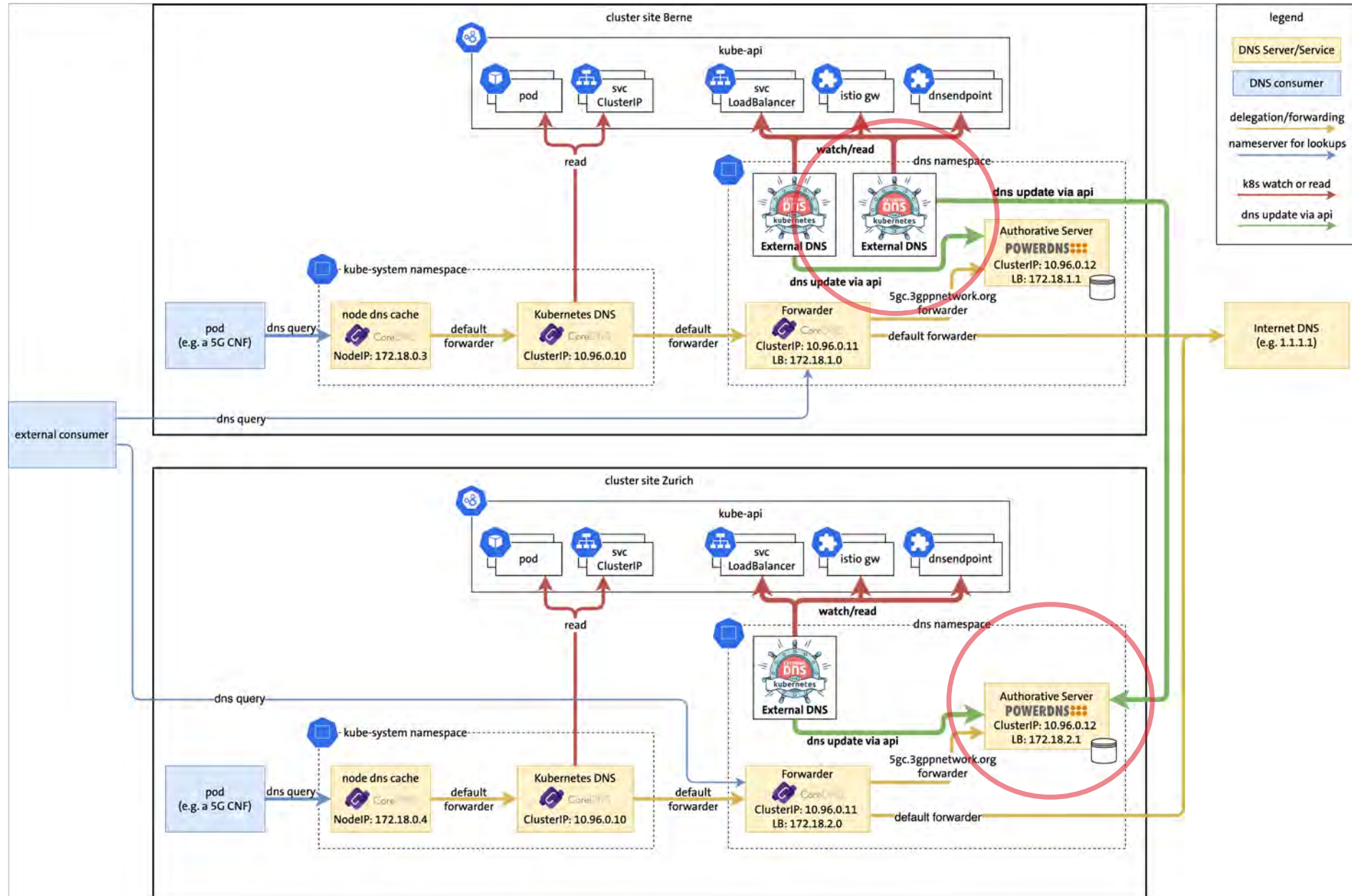


Dual-Cluster Using ExternalDNS



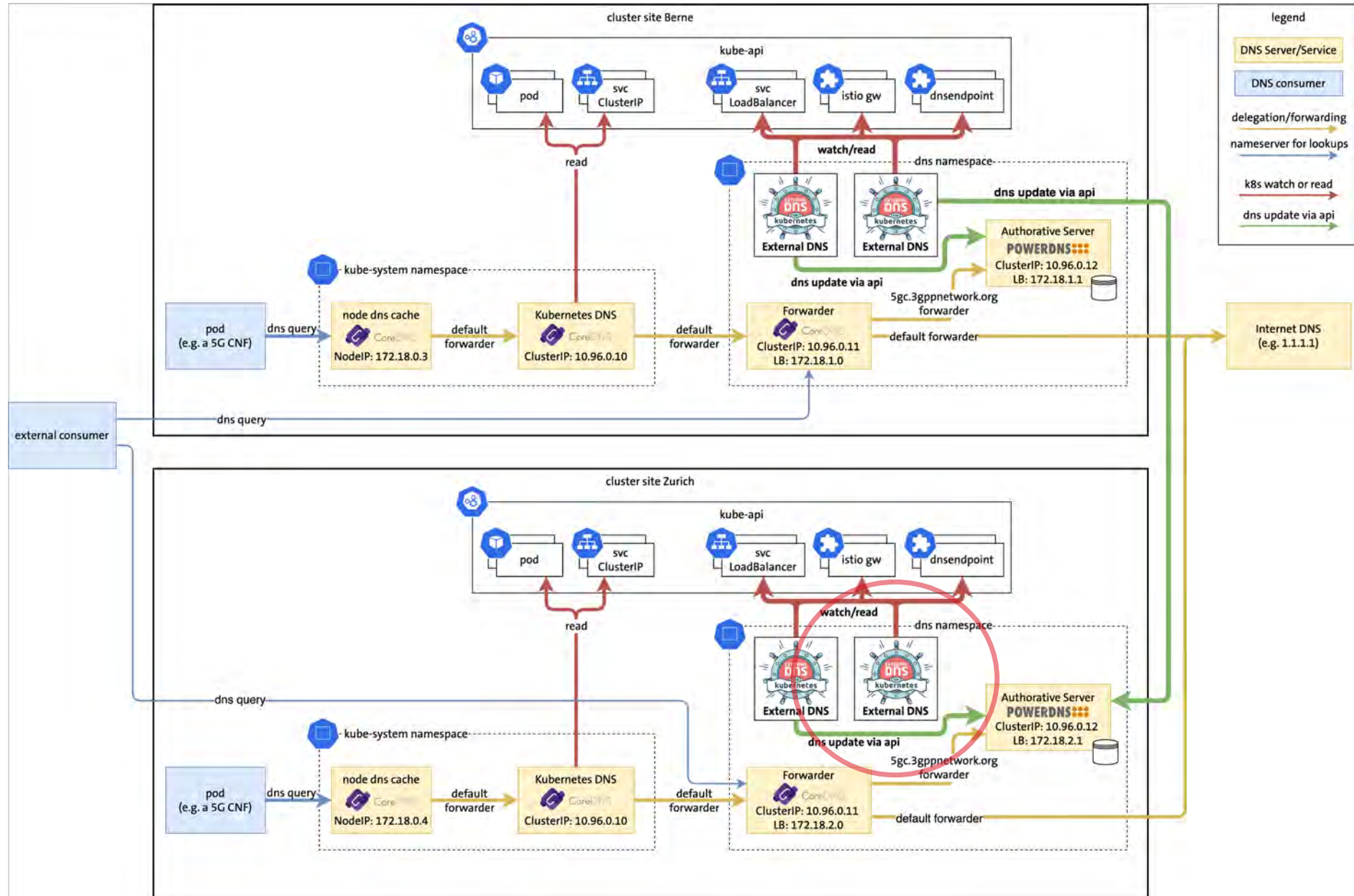


Dual-Cluster Using ExternalDNS



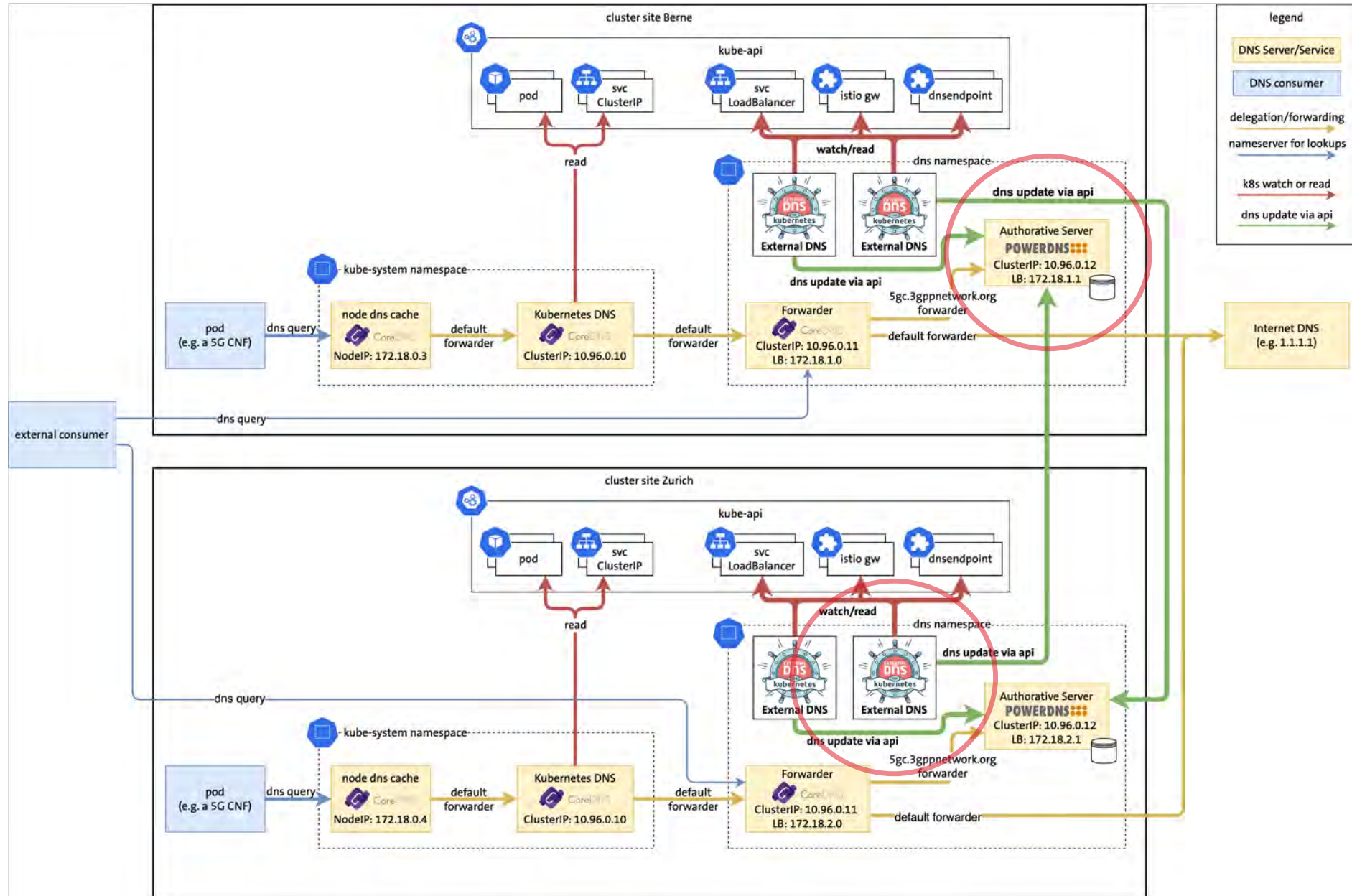


Dual-Cluster Using ExternalDNS



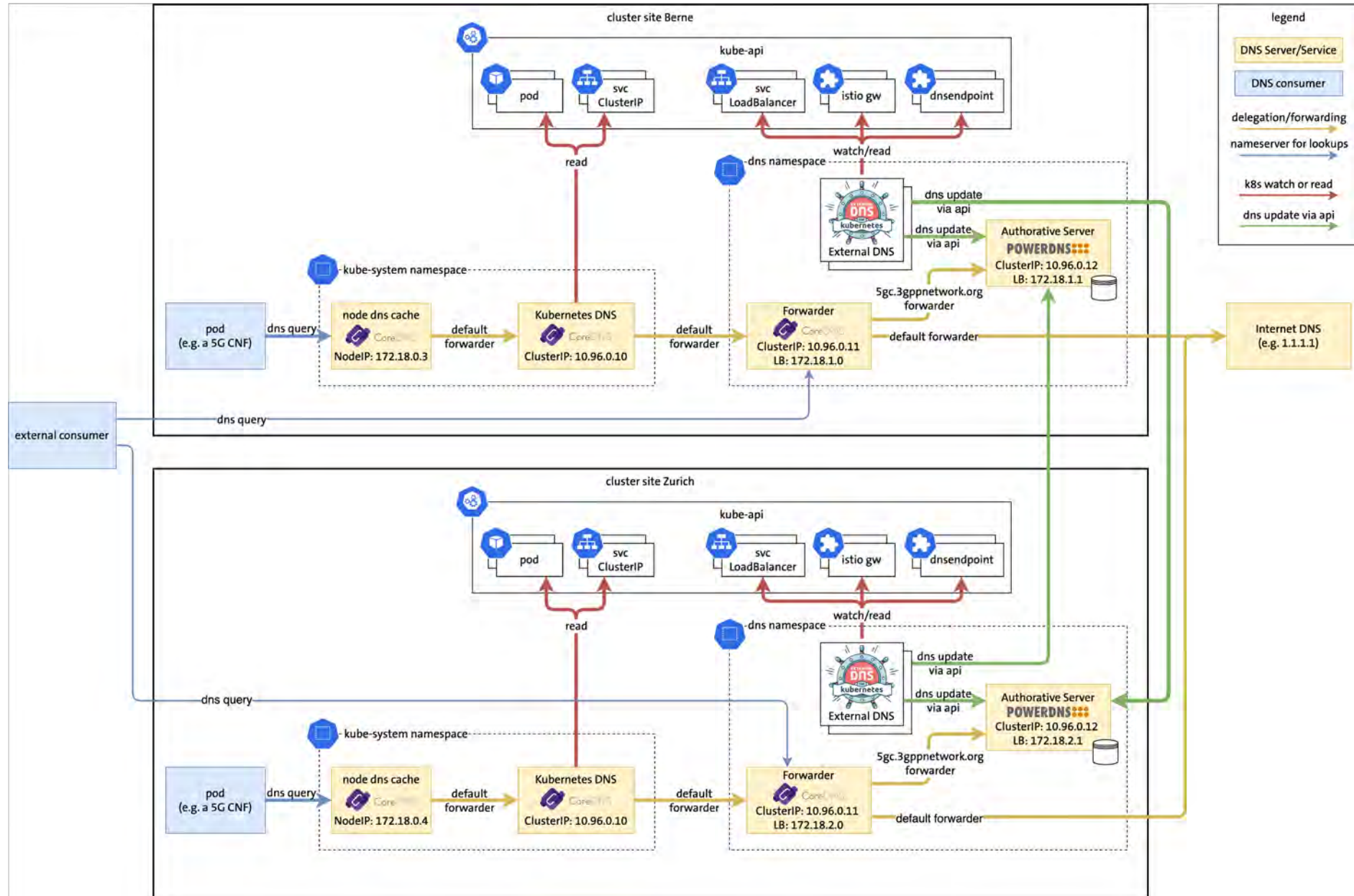


Dual-Cluster Using ExternalDNS



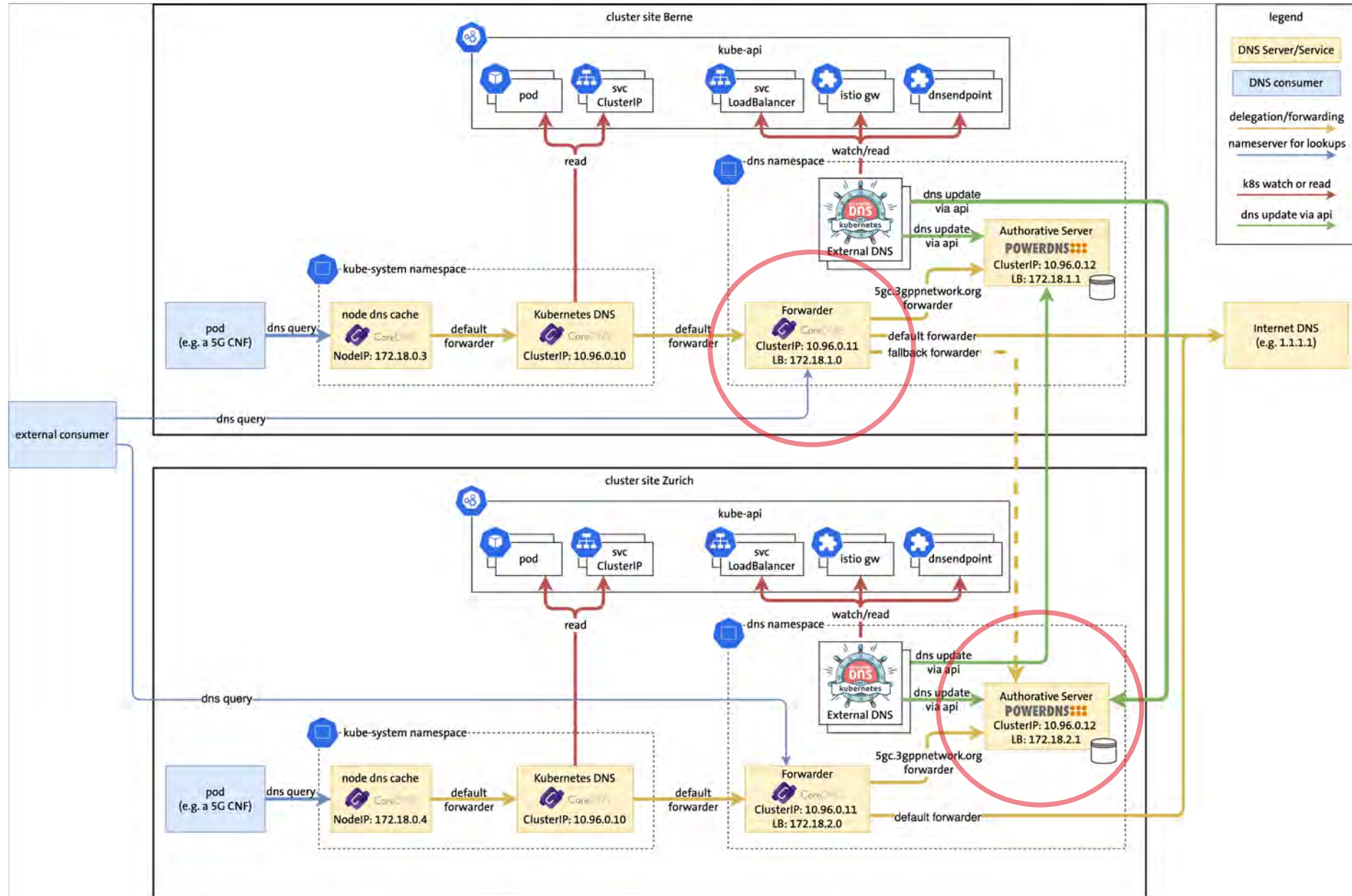


Dual-Cluster Using ExternalDNS



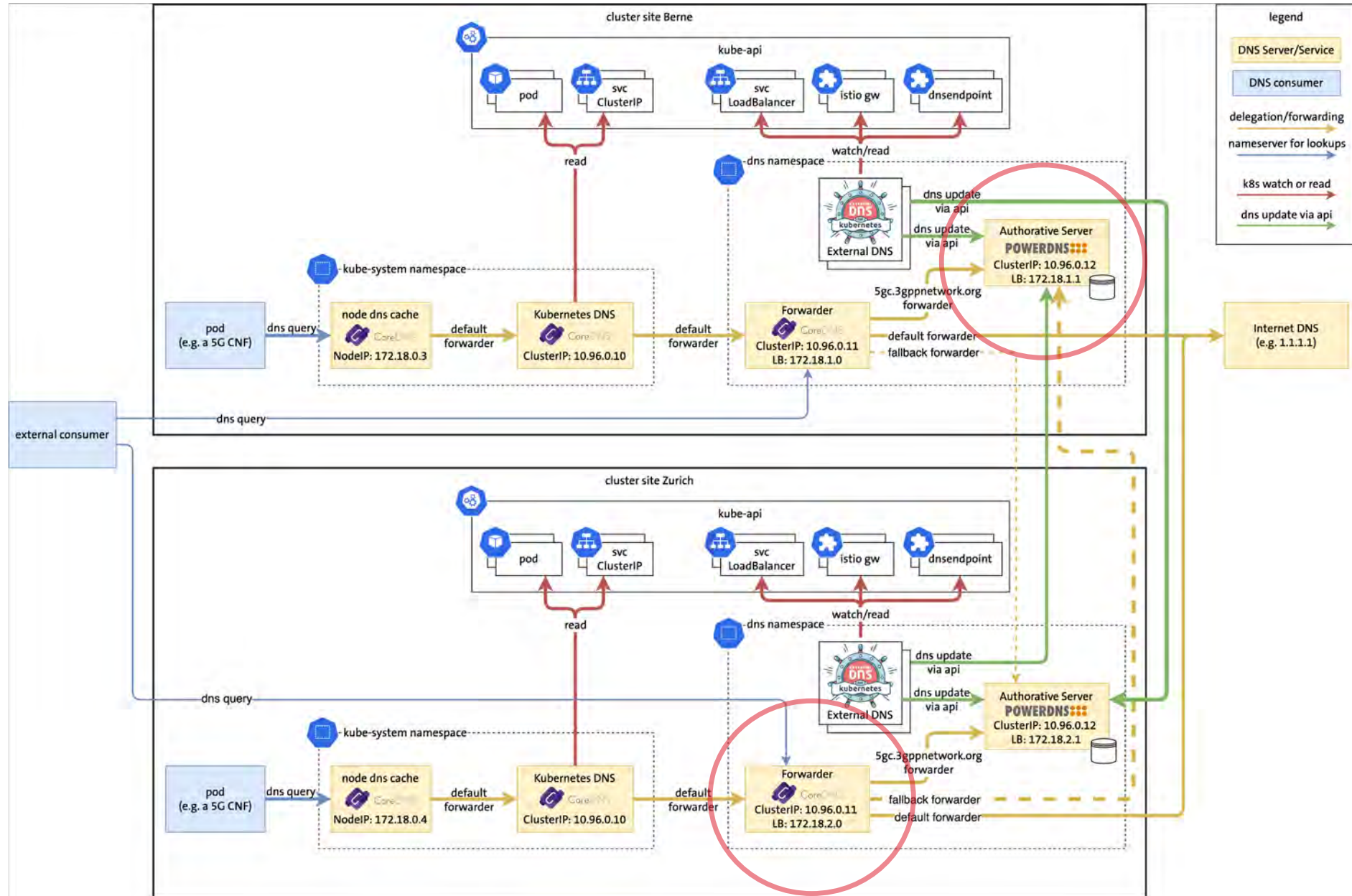


Eliminating Single Point of Failure



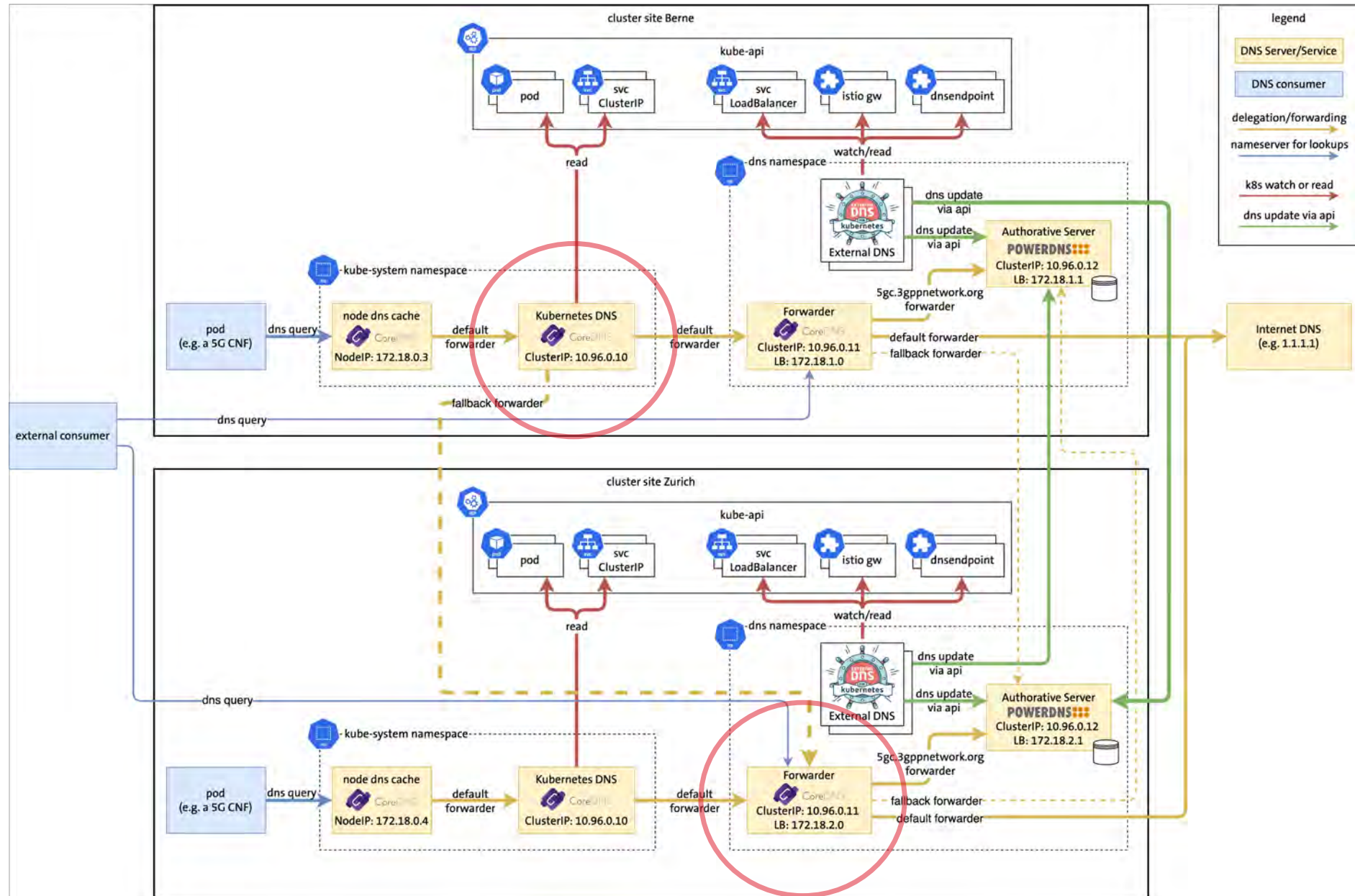


Eliminating Single Point of Failure



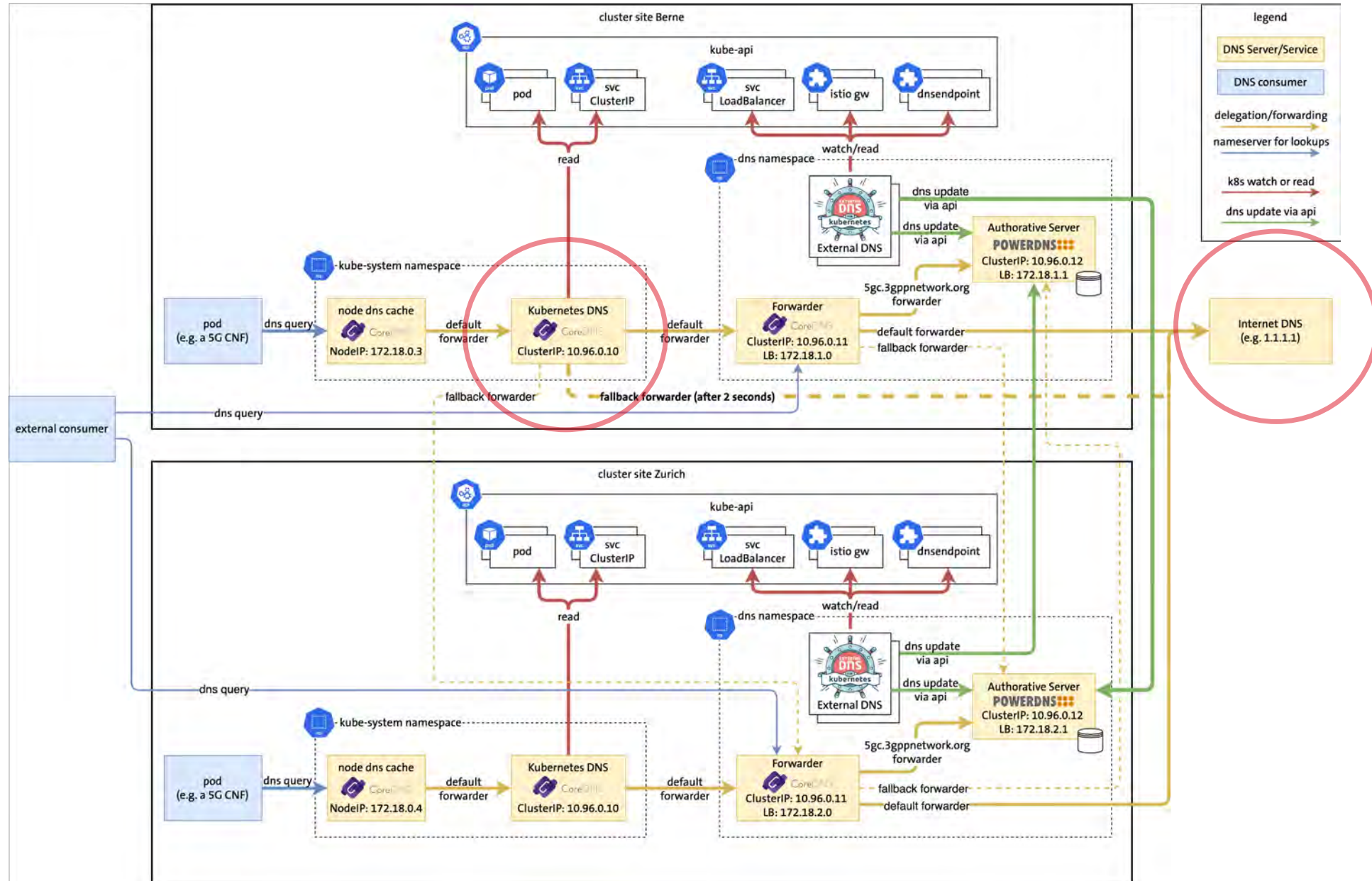


Eliminating Single Point of Failure



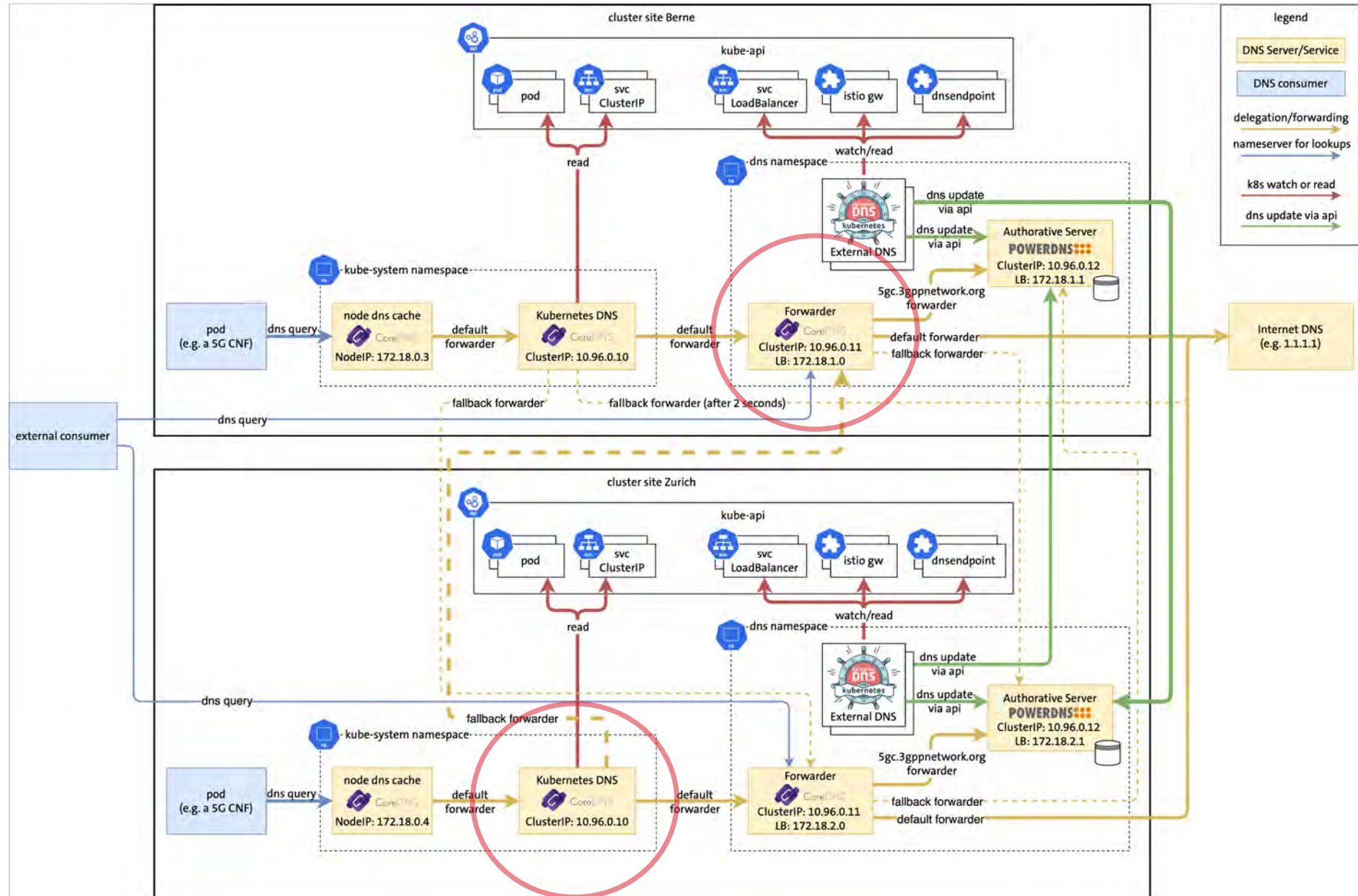


Eliminating Single Point of Failure



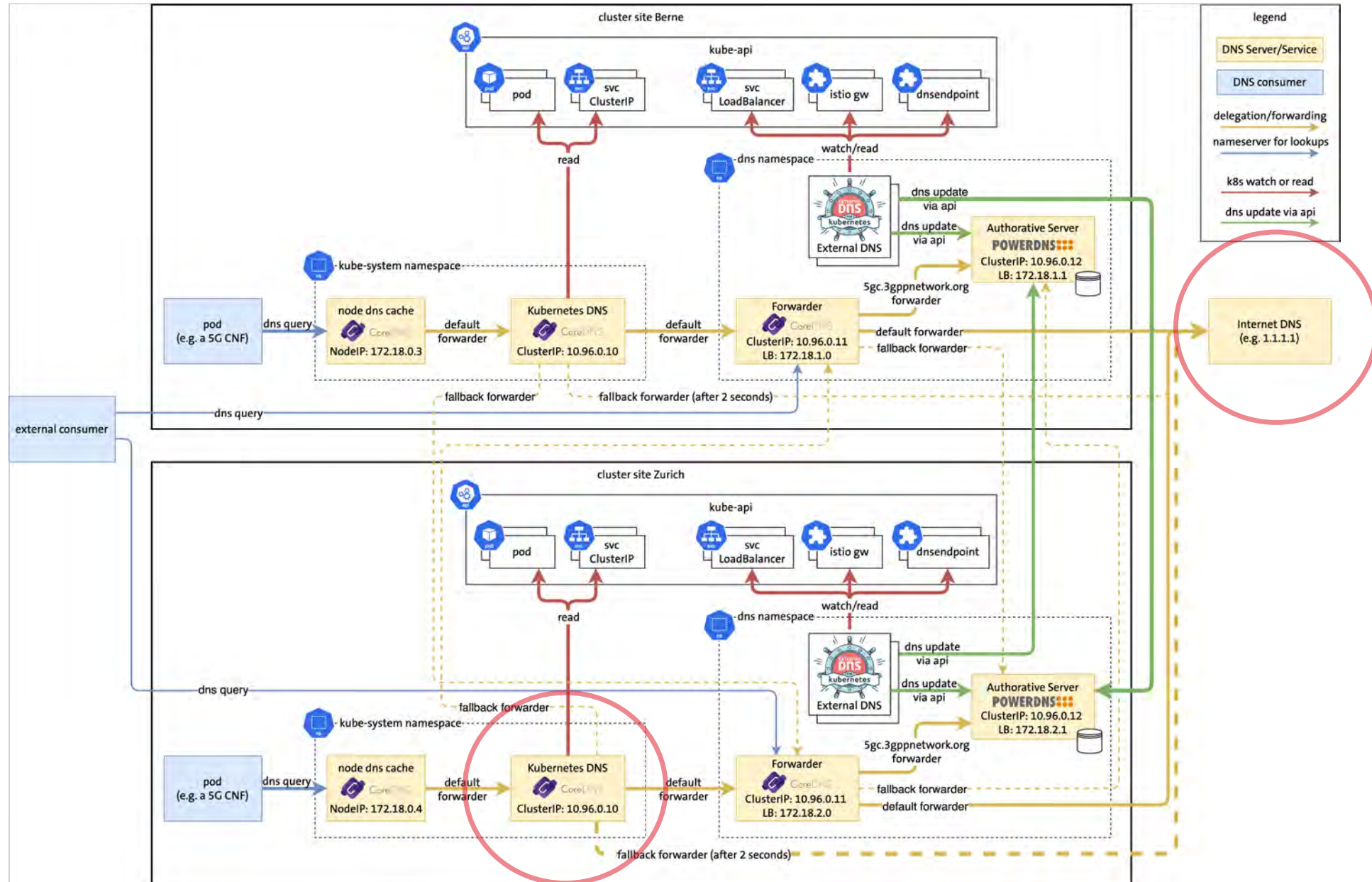


Eliminating Single Point of Failure



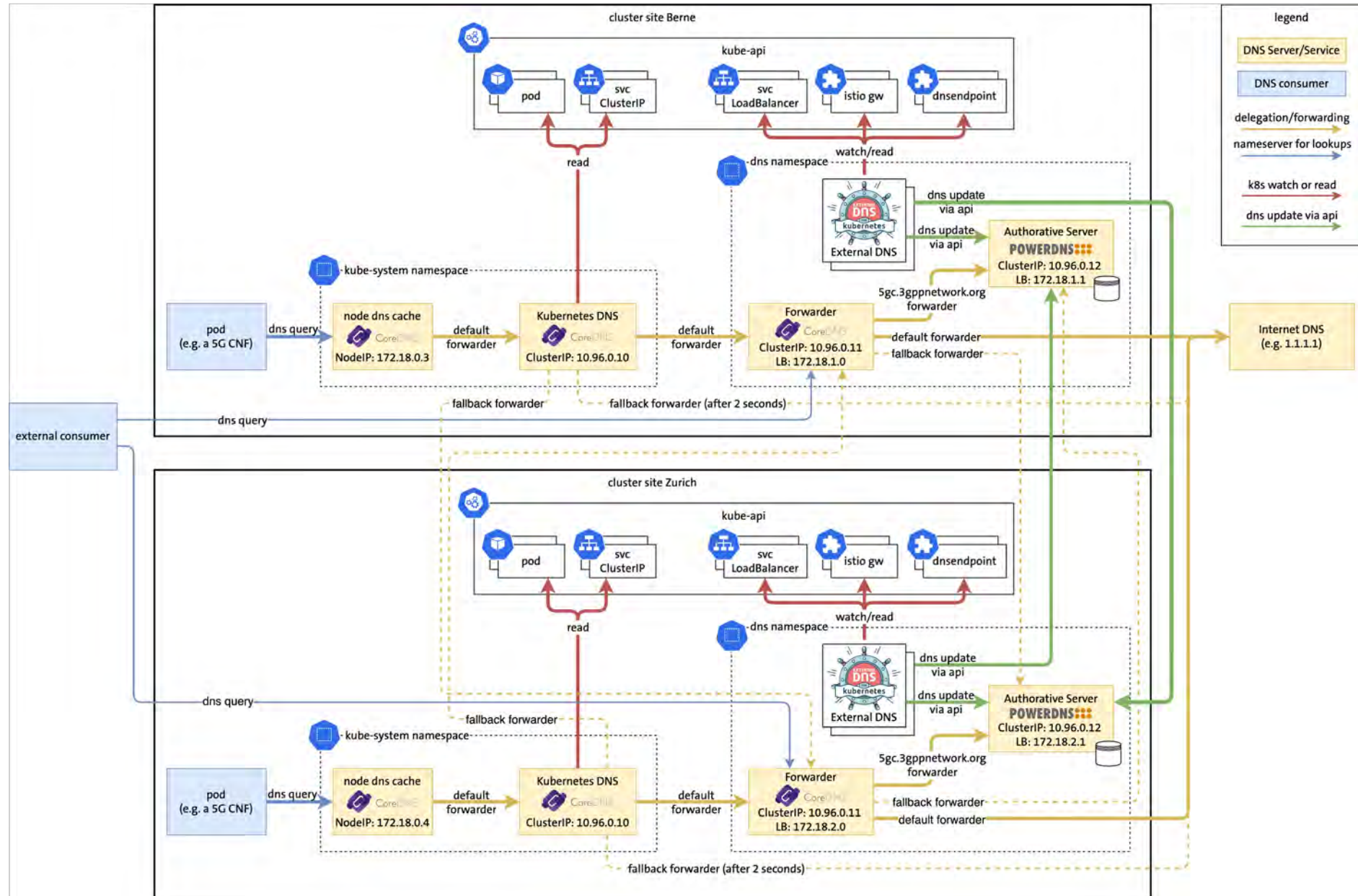


Eliminating Single Point of Failure





Final Dual Cluster Setup





In case that was too fast...

Designing and Evaluating a Resilience Testing Strategy for Cloud-Native DNS Systems

by Fabian Schulz & Joel Studler at Open Source Summit Europe 2025

<https://osseu2025.sched.com/event/25VmB/dig-smart-building-a-reliable-cloud-native-dns-for-modern-networks>

Building and Operating a Highly Reliable Cloud Native DNS Service With Open Source Technologies

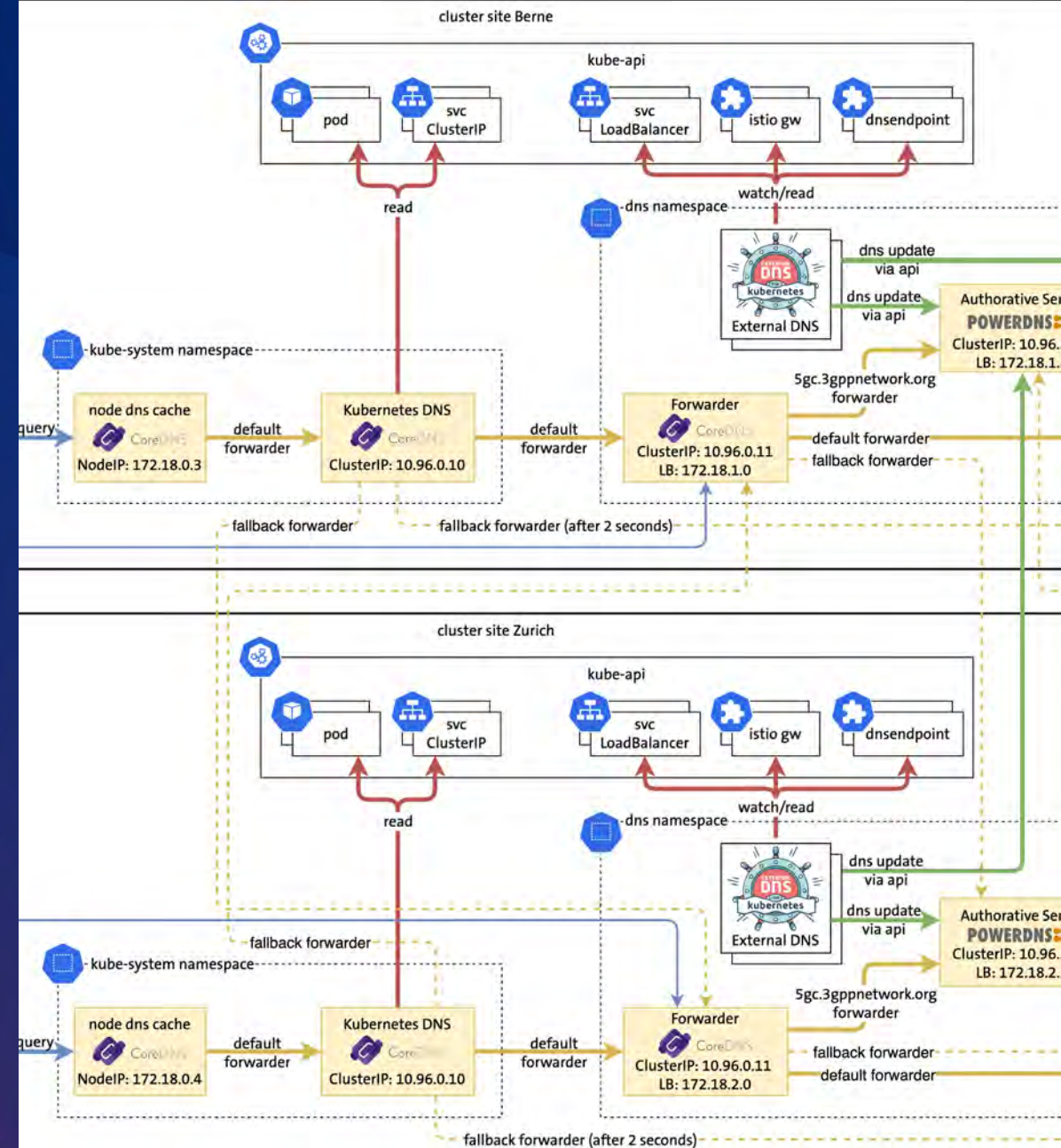
by Fabian Schulz & Hoang Anh Mai, Joel Studler at ContainerDays 2024

<https://www.youtube.com/watch?v=XGdGgSKadtw>

<https://github.com/swisscom/cloud-native-telco/>



Demo Multi Cluster





Traffic Generation
Record Updates
Accuracy Estimation
50 QPS

Monitoring Prometheus & Grafana QPS , Latency, Accuracy

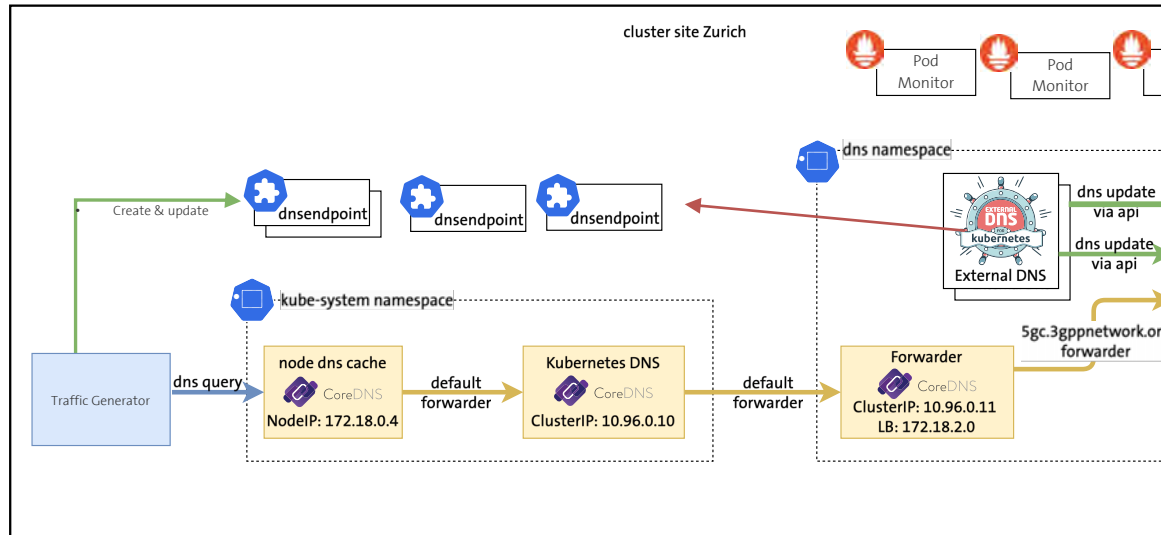
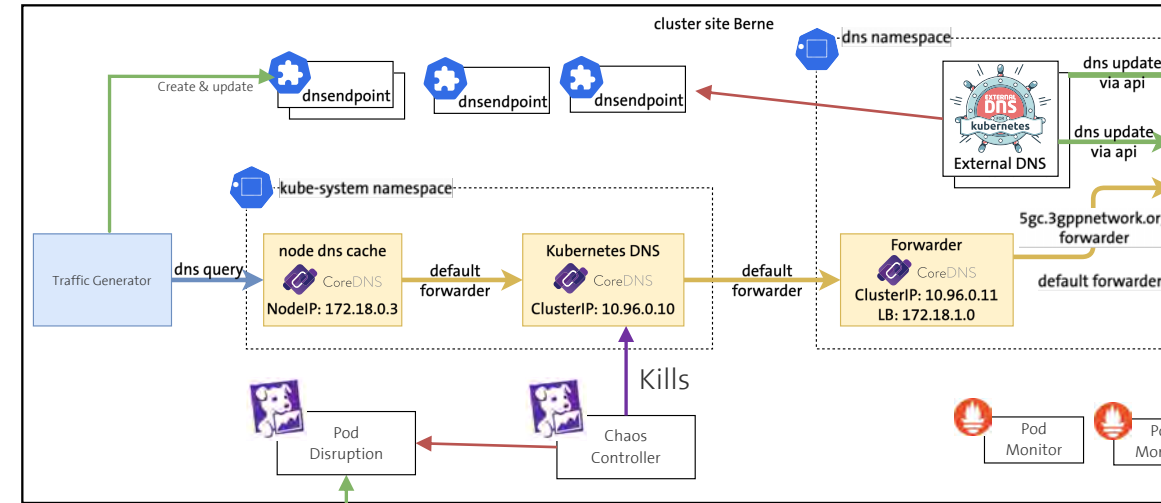
Chaos Injection

Deployment scale down

- Forwarder
- Power DNS



Demo Resilience Testing



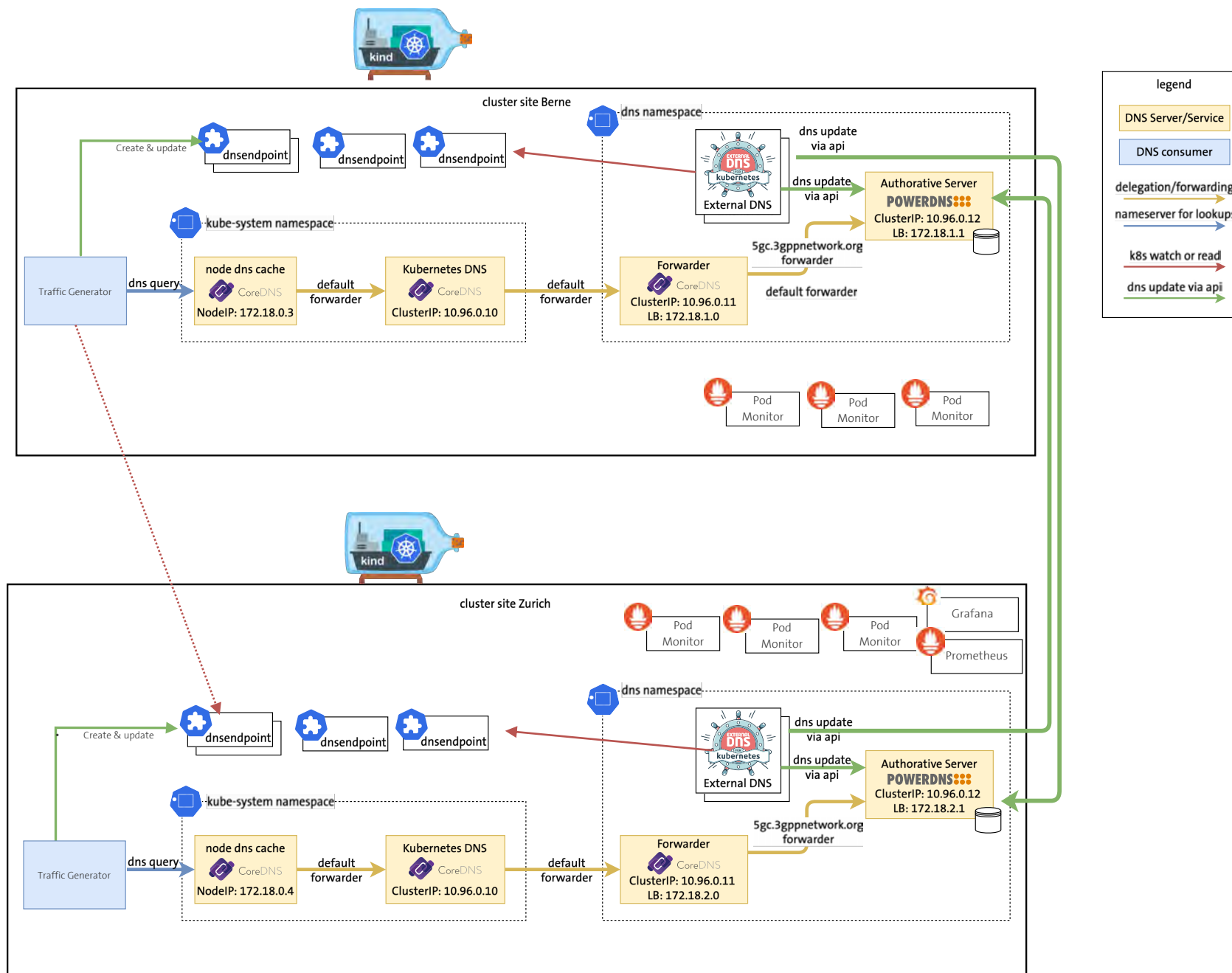


Limitations

Accuracy Cross Cluster

No Record Deletions

Kind Clusters





What Did We Achieve?



Geo Redundant & HA

Spread across multiple K8s clusters and geo regions to increase reliability

✓ Spread across multiple K8s Clusters



Fully Automated

GitOps driven and automated provisioning of DNS records

✓ GitOps + ExternalDNS



Proximity to Consumer

Minimal amount of hops between 5G Core and DNS

✓ On-prem deployment



Minimal Amount of SPOFs

Remove single points of failure from the System

✓ Distributed control plane
✓ Cross-Cluster Forwarding



K8s integration with ExternalDNS

The System leverages Kubernetes Patterns such as CRs and Operators

✓ 100% Kubernetes Resources



Support of Advanced DNS features

Resource Records such as NAPTR and SRV supported for e.g. SIP Phone Calls

✓ Advanced RRs supported



Context & Related Resources

Designing and Evaluating a Resilience Testing Strategy for Cloud-Native DNS Systems

Peer reviewed publication by Georgios Daskalopoulos; Thomas Grübl; Burkhard Stiller

<https://ieeexplore.ieee.org/abstract/document/11579218>

<https://dl.ifip.org/db/conf/networking/networking2026/1571261858.pdf>

Evolving GitOps: Harnessing Kubernetes Resource Model for 5G

Talk by Ashan Senevirathne and Joel Studler at Open Source Summit 2024

https://www.youtube.com/watch?v=35-fE_gHDjw

From Weeks to Seconds: Scaling Network Automation with Kubernetes Operators

Talk by Alessio Diamanti and Adrian Kurt at ContainerDays 2025

<https://www.youtube.com/watch?v=RytDzZYDQTU>

<https://github.com/swisscom/cloud-native-telco/>



Thank You!



Code/Slides Q&A